

Big-Data-Assignment2 — search engine (MapReduce + Cassandra + Spark)

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Methodology

The project implements a batch retrieval pipeline on Hadoop and Cassandra, packaged with Docker Compose.

Data preparation. A PySpark job runs on YARN via spark-submit. The source is either a.parquet on HDFS (up to N_DOCS documents, default 100) or plain files under app/data/*.txt. The job writes HDFS /data (one file per document) and /input/data (a single dataset of tab-separated id, title, body lines). Parquet text columns are cleaned when needed; /data filenames are derived from id and title. Spark coalesce and saveAsTextFile produce one merged input file for MapReduce.

Indexing. Hadoop MapReduce Streaming runs a mapper that tokenizes title and body (lowercase alphanumeric tokens), emits document-length statistics and postings (term, within-document frequency, document length, title). A single reducer aggregates global STATS (document count and total length), VOCAB (term → document frequency), and POSTING lines. Output is stored under /indexer/index/ on HDFS.

Cassandra load. A loader reads the MapReduce output from HDFS and fills keyspace search_engine: global corpus metrics for BM25 average length, vocabulary (df per term), and postings (term, doc id, tf, dl, title). Queries only touch rows for terms that appear in the query string.

Search. The query program reuses the same tokenizer as the indexer, fetches postings and df from Cassandra, computes BM25 (parameters k1, b from the environment), sums scores per document, and prints the top 10 doc_id and title lines. search.sh runs spark-submit with PySpark RDD aggregation; the text passed to the driver is set in BM25_QUERY. With SEARCH_USE_SPARK=0, the same scoring runs in-process in Python with the query on stdin.

Infrastructure. start-services.sh starts HDFS and YARN. app.sh waits for safe mode, creates a Python venv (under /tmp/search_venv, packaged as /app/.venv.tar.gz for workers), installs requirements.txt, then runs preparation → index.sh (MapReduce + Cassandra). A sample search runs at the end; app.sh uses local Spark for that step so the container exits promptly. A separate search.sh invocation defaults to YARN for cluster execution.

Demonstration

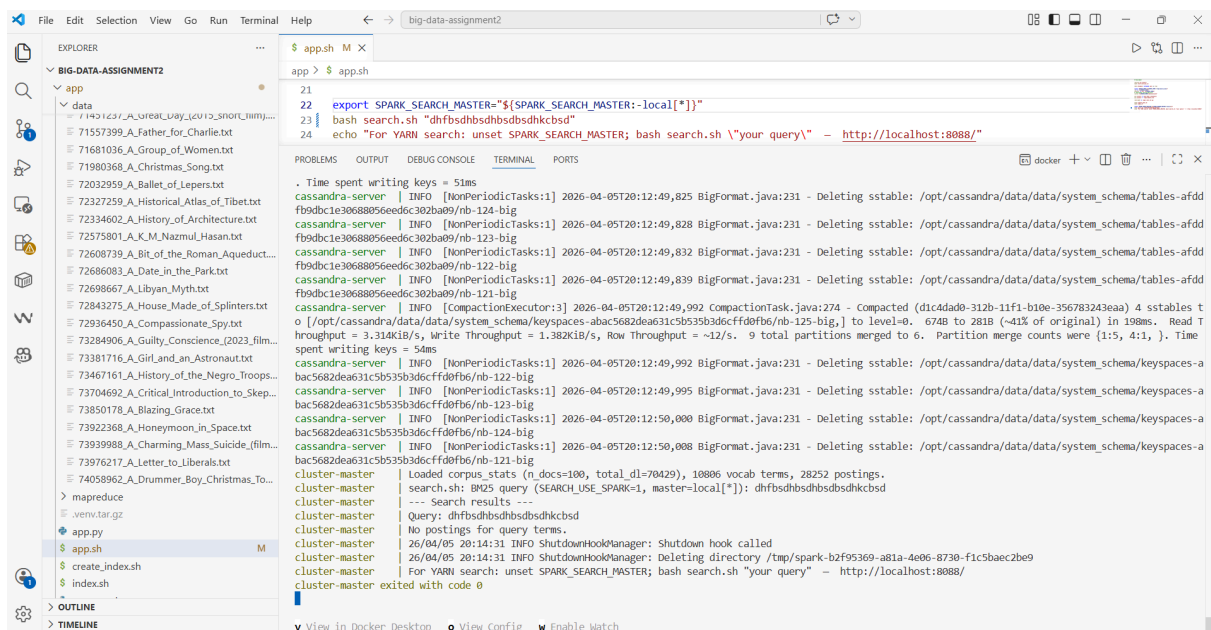
How to run. Install Docker and Docker Compose. From the repository root:

```
| docker compose up
```

Results. High ranks go to documents with strong within-document term frequency for query terms and comparatively rare terms in the collection (higher IDF). Document length is normalized by BM25. On a small corpus, different queries that share frequent words may yield similar top lists.

- Synthetic query with random characters

The engine correctly reports no postings when query terms are absent from the vocabulary.

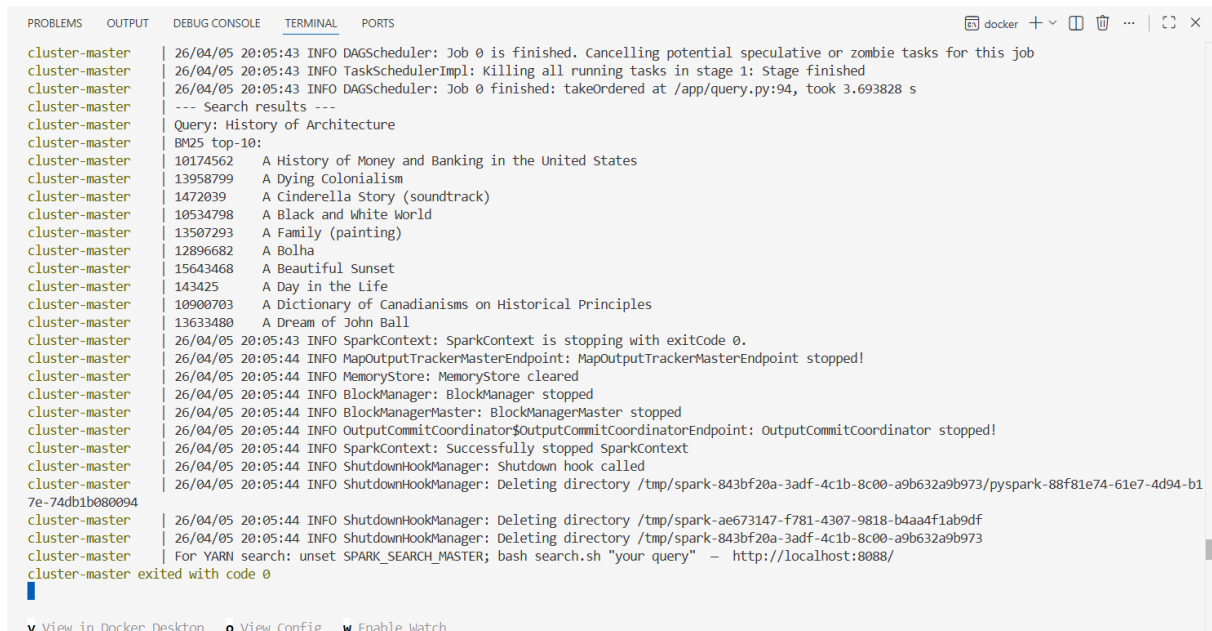
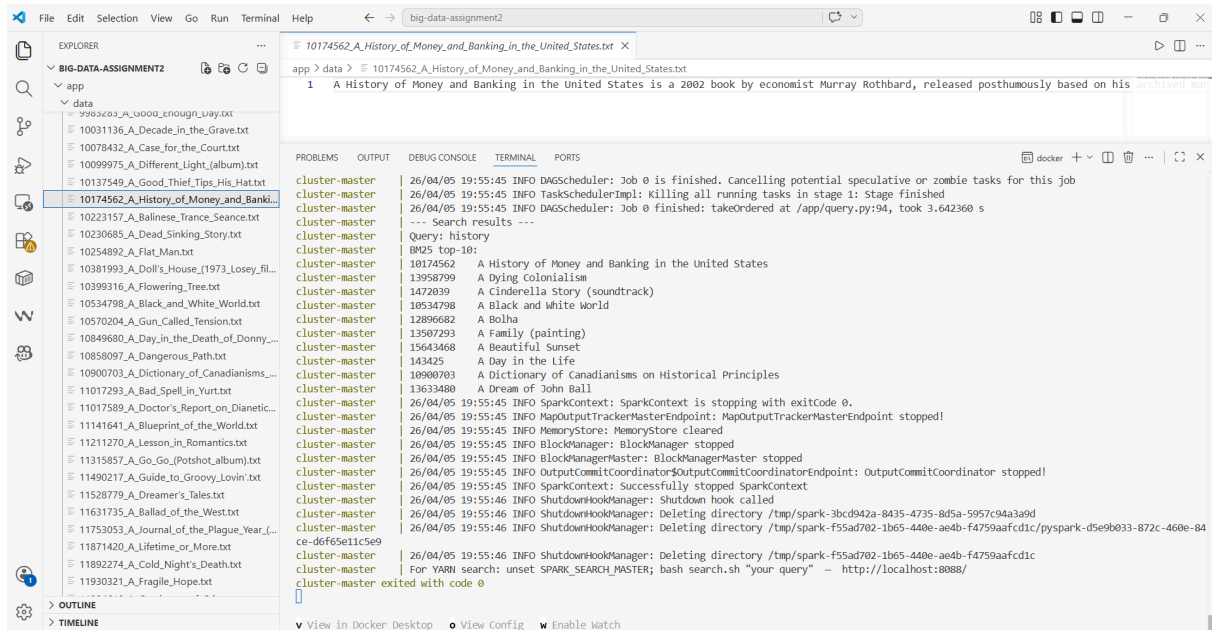


```
$ app.sh
app > $ app.sh
21
22 export SPARK_SEARCH_MASTER="${SPARK_SEARCH_MASTER:-local[*]}"
23 bash search.sh "dhfbsdhbsdhbsdhkcbdsd"
24 echo "For YARN search: unset SPARK_SEARCH_MASTER; bash search.sh \"your query\" - http://localhost:8088/"

. Time spent writing keys = 51ms
cassandra-server | INFO [NonPeriodicTasks:1] 2026-04-05T20:12:49,825 BigFormat.java:231 - Deleting sstable: /opt/cassandra/data/data/system_schema/tables-afdd
fb9dbc1e3068805eed6c302ba09/nb-124-big
cassandra-server | INFO [NonPeriodicTasks:1] 2026-04-05T20:12:49,828 BigFormat.java:231 - Deleting sstable: /opt/cassandra/data/data/system_schema/tables-afdd
fb9dbc1e3068805eed6c302ba09/nb-123-big
cassandra-server | INFO [NonPeriodicTasks:1] 2026-04-05T20:12:49,832 BigFormat.java:231 - Deleting sstable: /opt/cassandra/data/data/system_schema/tables-afdd
fb9dbc1e3068805eed6c302ba09/nb-122-big
cassandra-server | INFO [NonPeriodicTasks:1] 2026-04-05T20:12:49,839 BigFormat.java:231 - Deleting sstable: /opt/cassandra/data/data/system_schema/tables-afdd
fb9dbc1e3068805eed6c302ba09/nb-121-big
cassandra-server | INFO [CompactionExecutor:3] 2026-04-05T20:12:49,992 CompactionTask.java:274 - Compacted (d1c4da0-312b-11f1-b10e-356783243eaa) 4 sstables t
o [/opt/cassandra/data/data/system_schema/keyspaces-abac5682dea631c5b535b3dcffdf0fb6/nb-125-big,] to level=0. 6748 to 281B (~41% of original) in 198ms. Read T
hroughput = 3.314KiB/s, Write Throughput = 1.382KiB/s, Row Throughput = ~12/s. 9 total partitions merged to 6. Partition merge counts were {1:5, 4:1, }. Time
spent writing keys = 54ms
cassandra-server | INFO [NonPeriodicTasks:1] 2026-04-05T20:12:49,992 BigFormat.java:231 - Deleting sstable: /opt/cassandra/data/data/system_schema/keyspaces-a
bac5682dea631c5b535b3dcffdf0fb6/nb-122-big
cassandra-server | INFO [NonPeriodicTasks:1] 2026-04-05T20:12:49,995 BigFormat.java:231 - Deleting sstable: /opt/cassandra/data/data/system_schema/keyspaces-a
bac5682dea631c5b535b3dcffdf0fb6/nb-123-big
cassandra-server | INFO [NonPeriodicTasks:1] 2026-04-05T20:12:50,000 BigFormat.java:231 - Deleting sstable: /opt/cassandra/data/data/system_schema/keyspaces-a
bac5682dea631c5b535b3dcffdf0fb6/nb-124-big
cassandra-server | INFO [NonPeriodicTasks:1] 2026-04-05T20:12:50,008 BigFormat.java:231 - Deleting sstable: /opt/cassandra/data/data/system_schema/keyspaces-a
bac5682dea631c5b535b3dcffdf0fb6/nb-121-big
cluster-master | Loaded corpus_stats (n_docs=100, total_d=70429), 10806 vocab terms, 28252 postings.
cluster-master | search.sh: BM25 query (SEARCH_USE_SPARK=1, master=local[*]): dhfbsdhbsdhbsdhkcbdsd
cluster-master | --- Search results ---
cluster-master | Query: dhfbsdhbsdhbsdhkcbdsd
cluster-master | No postings for query terms.
cluster-master | 26/04/05 20:14:31 INFO ShutdownHookManager: Shutdown hook called
cluster-master | 26/04/05 20:14:31 INFO ShutdownHookManager: Deleting directory /tmp/spark-b2f95369-a81a-4e06-8730-f1c5baec2be9
cluster-master | For YARN search: unset SPARK_SEARCH_MASTER; bash search.sh "your query" - http://localhost:8088/
cluster-master exited with code 0
```

- Similar ranked lists for queries that share the same key terms

Both queries share high-weight tokens such as history, so BM25 largely ranks the same documents; the extra token architecture barely changes the scores if it is rare or absent in the small corpus.



- Different queries

Two distinct queries produce different top-10 rankings, which confirms the search pipeline is working as expected.

